

Redox & Electrochemistry

Complete Study Notes

Definitions · oxidation numbers · half-equations · galvanic & electrolytic cells · standard electrode potentials · Faraday calculations · corrosion · redox titrations · exam traps · full data sheet.

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1 Redox Fundamentals

A **redox reaction** involves the transfer of electrons between species. **Oxidation** and **reduction** always happen together — the electrons lost by one species are gained by another. Electrons are never created or destroyed.

OIL RIG

Oxidation Is Loss · Reduction Is Gain (of electrons)

Process	Electrons	Oxidation number	Oxygen	Hydrogen
Oxidation	lost	increases	gained	lost
Reduction	gained	decreases	lost	gained

The electron-transfer definition is the most general and is the one to use at ATAR level. The O/H definitions are handy shortcuts for combustion and organic reactions.

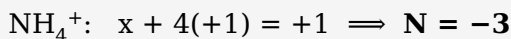
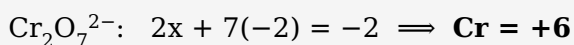
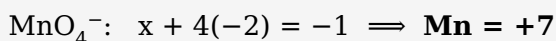
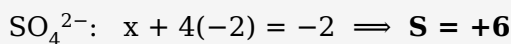
2 Oxidation Numbers (States)

The **oxidation number** is the charge an atom *would* have if every bond were fully ionic. It is the bookkeeping tool for tracking electron transfer and identifying what is oxidised or reduced.

RULES — APPLY IN THIS PRIORITY ORDER

- Free element = **0** (e.g. Na, O₂, Cl₂, S₈, P₄).
- Monatomic ion = its **charge** (Na⁺=+1, S²⁻=−2, Al³⁺=+3).
- Group 1 = +1; Group 2 = +2; Al = +3 (in compounds).
- F = −1 always. H = +1 (but −1 in metal hydrides, e.g. NaH).
- O = −2 (but −1 in peroxides e.g. H₂O₂; +2 in OF₂).
- Sum = **0** in a neutral compound; = **ion charge** in a polyatomic ion.

WORKED EXAMPLES — SOLVE FOR THE UNKNOWN



USING OXIDATION NUMBERS TO IDENTIFY REDOX

Compare each element before and after: an **increase** = oxidised; a **decrease** = reduced; **no change** = not a redox reaction (e.g. most acid-base and precipitation reactions).

Disproportionation

The *same* element is simultaneously oxidised *and* reduced:



3 Oxidants, Reductants & Conjugate Pairs

DEFINITIONS

- **Oxidising agent (oxidant):** accepts electrons $\rightarrow$ is itself **reduced**.
- **Reducing agent (reductant):** donates electrons $\rightarrow$ is itself **oxidised**.

COMMON CONFUSION

The oxidising agent is the species that gets **reduced**; the reducing agent is the species that gets **oxidised**. Read it twice.

Each pair is linked by electron transfer — a **conjugate redox pair**:



e.g. Cu^{2+}/Cu and Zn^{2+}/Zn . The stronger an oxidant, the weaker its conjugate reductant.

Common species at ATAR

Strong oxidising agents	Strong reducing agents
F_2 , $\text{MnO}_4^-/\text{H}^+$, $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$, Cl_2 , H_2O_2 , conc. HNO_3	Reactive metals (Li, K, Na, Mg, Al, Zn), H_2 , C, I^- , SO_2

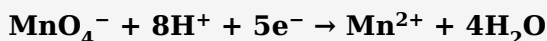
4 Writing & Balancing Half-Equations

A **half-equation** shows one process (oxidation or reduction) with its electrons. Electrons appear on the **left** of a reduction and on the **right** of an oxidation.

BALANCING IN ACIDIC SOLUTION — 5 STEPS

1. Balance all atoms **except O and H**.
2. Balance **O** with H_2O .
3. Balance **H** with H^+ .
4. Balance **charge** with e^- .
5. Check atoms *and* charge.

EXAMPLE — PERMANGANATE REDUCED IN ACID



BASIC (ALKALINE) SOLUTION

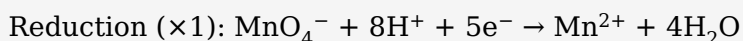
Balance as if acidic, then add **OH⁻** to both sides equal to the H⁺ present. Combine H⁺ + OH⁻ → H₂O and cancel any water appearing on both sides.

5 Combining Half-Equations

METHOD

1. Write the oxidation and reduction half-equations.
2. Multiply each so the **electrons are equal**.
3. Add them; **cancel electrons** and any identical species on both sides.

EXAMPLE — MnO_4^- OXIDISING Fe^{2+} IN ACID



Charge check: left $(-1+8+10)=+17$; right $(+2+15)=+17$ ✓

6 Galvanic (Voltaic) Cells

A galvanic cell converts **chemical** → **electrical** energy using a **spontaneous** redox reaction ($E^\circ_{\text{cell}} > 0$). Separating the two half-reactions forces electrons through an external wire — this is the current.

Component	Role
Anode	Oxidation occurs here. Galvanic → negative terminal.
Cathode	Reduction occurs here. Galvanic → positive terminal.
Salt bridge	Inert electrolyte (e.g. KNO_3); completes the circuit and keeps each half-cell electrically neutral by allowing ion flow. It does <i>not</i> carry electrons.
External wire	Carries electrons anode → cathode; holds the voltmeter/load.

MEMORY AIDS

- **AN OX** = ANode is OXidation. **RED CAT** = REDuction is CATHode.
- Electrons always flow **anode** → **cathode** in the wire.
- In the salt bridge: **anions** → **anode**, **cations** → **cathode**.
- Galvanic polarity: anode **-**, cathode **+** (reversed in electrolysis).

DANIELL CELL (Zn | Cu)

Anode: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$ ($E^\circ = -0.76 \text{ V}$)

Cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$ ($E^\circ = +0.34 \text{ V}$)

Overall: $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$ $E^\circ_{\text{cell}} = +1.10 \text{ V}$

Zn electrode loses mass; Cu electrode gains mass; $[\text{Cu}^{2+}]$ falls, $[\text{Zn}^{2+}]$ rises.

Cell diagram notation

Anode on the left. Single line | = phase boundary; double line || = salt bridge.



If a half-cell has no solid conductor (e.g. $\text{Fe}^{3+}/\text{Fe}^{2+}$) use an **inert electrode** (Pt or graphite): $\text{Pt} \mid \text{Fe}^{2+}, \text{Fe}^{3+}$.

7 Standard Electrode Potentials

Each half-cell's tendency to gain electrons is its **standard reduction potential (E°)**, measured in volts against the **Standard Hydrogen Electrode (SHE)**, defined as exactly **0.00 V**.

STANDARD CONDITIONS (THE "°")

Solutions **1.0 mol L⁻¹** · gases **100 kPa** · temperature **25 °C (298 K)** · measured vs the SHE.

READING THE TABLE (ALL VALUES ARE REDUCTIONS)

- **More positive E°** → stronger **oxidising agent** (top-left of table; e.g. F_2).
- **More negative E°** → its reduced form is a stronger **reducing agent** (bottom-right; e.g. Li).

PREDICTION RULE ("ANTICLOCKWISE")

Place the two half-equations with the more positive E° on top. The **top reaction goes forward (reduction)** and the **bottom reverses (oxidation)**: the oxidant (top-left) reacts with the reductant (bottom-right).

8 Cell EMF & Predicting Reactions

STANDARD CELL POTENTIAL

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

Use both values straight from the table as reductions — the minus sign handles the reversal; do **not** flip the sign yourself.

SIGN OF E°_{cell}

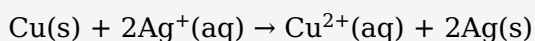
- $E^\circ_{\text{cell}} > 0$ → spontaneous (works as a galvanic cell); $\Delta G < 0$.
- $E^\circ_{\text{cell}} < 0$ → non-spontaneous (needs electrolysis).

Free-energy link: $\Delta G^\circ = -nFE^\circ_{\text{cell}}$ ($n = \text{mol e}^-$, $F = 96\,500 \text{ C mol}^{-1}$).

EXAMPLE — DOES COPPER METAL REACT WITH Ag^+ ?

$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ (+0.80 V) is reduced (cathode); $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ is oxidised (anode).

$$E^\circ_{\text{cell}} = (+0.80) - (+0.34) = \mathbf{+0.46 \text{ V}} \rightarrow \text{spontaneous}$$



E° DOES NOT SCALE WITH AMOUNT

E° is **intensive**. When you multiply a half-equation to balance electrons, do **not** multiply its E° value.

Limitations of E° predictions

- Only valid at **standard conditions**; concentration, temperature and pressure shift the real potential.
- Predicts **feasibility, not rate** — a spontaneous reaction may be extremely slow.
- **Overpotential** can change the observed product (see §10).

9 Electrolytic Cells

Electrolysis uses an external power supply to drive a **non-spontaneous** redox reaction ($E^\circ_{\text{cell}} < 0$): electrical → chemical energy — the reverse of a galvanic cell.

Feature	Galvanic	Electrolytic
Energy	chemical → electrical	electrical → chemical
Spontaneity	spontaneous ($E^\circ > 0$)	non-spontaneous ($E^\circ < 0$)
Power supply	none (is the source)	required
Anode sign	negative (–)	positive (+)
Cathode sign	positive (+)	negative (–)

WHAT STAYS THE SAME

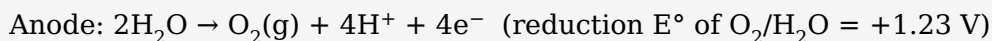
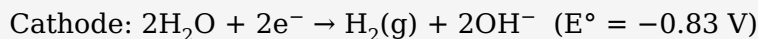
Oxidation is **always** at the anode and reduction **always** at the cathode; electrons **always** travel anode → cathode. Only the **polarity (+/–) flips** between cell types.

MOLTEN NaCl (DOWNS PROCESS)

No water present, so only the salt's ions react.

10 Predicting Electrolysis Products (Aqueous)

In water, **H₂O itself can be oxidised or reduced**, competing with the dissolved ions:

**DECISION PROCEDURE AT EACH ELECTRODE**

1. List every species that could react there (ions + water).
2. **Cathode:** the species with the **highest E°** is reduced.
3. **Anode:** the species with the **lowest (most negative) E°** is oxidised.
4. Then apply the practical factors below.

Factor	Effect on product
E° value	The primary predictor.
Concentration	A high ion concentration can cause it to discharge in preference to water (e.g. concentrated $\text{Cl}^- \rightarrow \text{Cl}_2$).

Overpotential	Extra voltage needed to release gases (especially O ₂); this is why Cl ₂ often forms instead of O ₂ .
Electrode type	Inert (Pt, C) is unchanged; an active electrode (e.g. Cu) may itself be oxidised at the anode.

EXAMPLE — AQUEOUS CuSO₄, INERT ELECTRODES

Cathode: Cu²⁺ (+0.34) vs H₂O (−0.83) → **Cu(s) deposited**

Anode: SO₄^{2−} very hard to oxidise → water oxidised → **O₂(g) + H⁺**

Copper plates out; O₂ bubbles at the anode; solution becomes acidic.

11 Faraday's Laws — Electrolysis Calculations

The amount of substance produced at an electrode is proportional to the charge passed.

FORMULAE & 5-STEP METHOD

Formula	Terms
$Q = I \times t$	Q charge (C); I current (A); t time (seconds)
$n(e^-) = Q \div F$	$F = 96\,500 \text{ C mol}^{-1}$
half-equation ratio	relate mol e [−] → mol product
$m = n \times M \quad / \quad V = n \times V_m$	convert to mass or gas volume

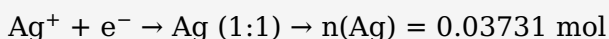
Steps: (1) $Q = It \rightarrow$ (2) $n(e^-) = Q/96\,500 \rightarrow$ (3) read electron:product ratio $\rightarrow$ (4) find mol product $\rightarrow$ (5) convert to mass/volume.

EXAMPLE A — SILVER ELECTROPLATING

2.00 A for 30.0 min through AgNO₃. Mass of Ag deposited?

$$Q = 2.00 \times (30.0 \times 60) = 3600 \text{ C}$$

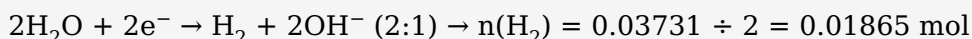
$$n(e^-) = 3600 \div 96\,500 = 0.03731 \text{ mol}$$



$$m = 0.03731 \times 107.9 = \mathbf{4.03 \text{ g Ag}}$$

EXAMPLE B — GAS VOLUME

Same 3600 C in electrolysis of water; V of H₂ at 25 °C, 100 kPa ($V_m = 24.79 \text{ L mol}^{-1}$)?



$$V = 0.01865 \times 24.79 = \mathbf{0.463 \text{ L (463 mL)}}$$

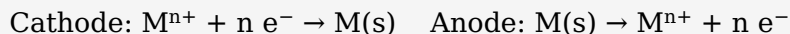
CALCULATION TRAPS

Convert time to **seconds**; watch the **electron ratio** (Cu²⁺ needs 2e[−], Al³⁺ needs 3e[−]); cells **in series** pass the **same charge**.

12 Industrial Applications

Electroplating

Object = **cathode**; pure plating metal = **anode**; electrolyte contains that metal's ions.

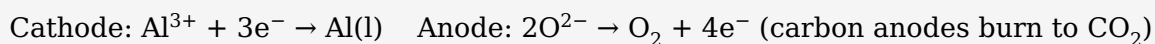


Electrorefining of copper

Impure Cu = anode (dissolves); pure Cu deposits on the cathode. Ag/Au fall as "anode sludge"; more reactive metals stay in solution.

Extraction of reactive metals (aluminium — Hall-Héroult)

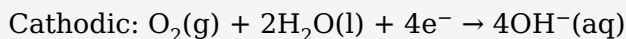
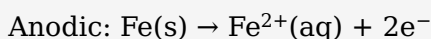
Very reactive metals (Al, Na, Mg, K) are obtained by electrolysis of the **molten** compound. Al_2O_3 is dissolved in molten cryolite to lower the melting point.



13 Corrosion of Iron & Its Prevention

Corrosion is unwanted oxidation of a metal. Rusting of iron is electrochemical and needs **both water and oxygen**; it is faster with dissolved salts (electrolytes) or acids.

THE ELECTROCHEMISTRY OF RUSTING



Fe^{2+} is further oxidised by O_2 to hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$ = rust.

Method	How it works
Barrier (paint, oil, plastic)	Excludes water/ O_2 ; fails if scratched.
Galvanising (Zn coat)	Barrier <i>plus</i> sacrificial protection — even if scratched, Zn oxidises before Fe.
Sacrificial anode (Zn/Mg block)	A more reactive metal is oxidised instead of the iron (ships, pipelines); replaced periodically.
Cathodic protection	External supply forces the iron to be the cathode , so it cannot oxidise.
Alloying (stainless steel)	Cr/Ni form a self-repairing protective oxide layer.

WHY SACRIFICIAL PROTECTION WORKS

A metal with a **more negative E°** than iron (Zn -0.76 , Mg -2.37 vs Fe -0.44) is a stronger reducing agent, so it is oxidised preferentially and protects the iron even where the coating breaks.

14 Redox Titrations

A redox titration finds an unknown concentration by reacting it completely with a standard oxidant/reductant, using the balanced equation's mole ratio.

PERMANGANATE TITRATIONS (SELF-INDICATING)

Acidified MnO_4^{-} (deep purple) $\rightarrow Mn^{2+}$ (colourless). The **endpoint** is the first **permanent faint pink** when one drop of excess MnO_4^{-} remains — no indicator needed.

WORKED EXAMPLE

25.0 mL of Fe^{2+} needs 20.0 mL of $0.0200 \text{ mol L}^{-1} \text{ KMnO}_4$. Find $[\text{Fe}^{2+}]$.

$$n(\text{MnO}_4^-) = 0.0200 \times 0.0200 = 4.00 \times 10^{-4} \text{ mol}$$

$$\text{Ratio } \text{MnO}_4^-:\text{Fe}^{2+} = 1:5 \rightarrow n(\text{Fe}^{2+}) = 2.00 \times 10^{-3} \text{ mol}$$

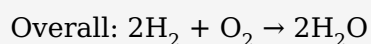
$$[\text{Fe}^{2+}] = 2.00 \times 10^{-3} \div 0.0250 = \mathbf{0.0800 \text{ mol L}^{-1}}$$

Iodine/thiosulfate titrations ($\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$) use a **starch** indicator; endpoint = blue-black just disappears.

15 Batteries & Fuel Cells

Type	Description & examples
Primary	Non-rechargeable; reaction not reversible (dry cell Zn/MnO_2 , alkaline).
Secondary	Rechargeable — recharging is electrolysis that reverses the reaction (lead-acid, lithium-ion).
Fuel cell	Reactants supplied continuously; high efficiency, low pollution (H_2/O_2).

HYDROGEN-OXYGEN FUEL CELL (ACIDIC)



16 Galvanic vs Electrolytic — Summary

	Galvanic (voltaic)	Electrolytic
Purpose	generates electricity	uses electricity to drive a reaction
Energy	chemical $\rightarrow$ electrical	electrical $\rightarrow$ chemical
$\Delta G / E^\circ_{\text{cell}}$	$\Delta G < 0$, $E^\circ > 0$ (spontaneous)	$\Delta G > 0$, $E^\circ < 0$ (non-spontaneous)
Power supply	none	required
Oxidation / Reduction	anode / cathode	anode / cathode
Anode sign	negative (–)	positive (+)
Cathode sign	positive (+)	negative (–)
Electron flow	anode $\rightarrow$ cathode	anode $\rightarrow$ cathode
Examples	batteries, fuel cells	electroplating, refining, Al/NaCl extraction, recharging

17 Exam Traps & Glossary

TOP TRAPS TO AVOID

- Oxidising agent is **reduced**; reducing agent is **oxidised**.
- Never change E° when multiplying a half-equation (E° is intensive).
- Salt bridge maintains **neutrality/completes the circuit** — it does not carry electrons.
- Only the $+/-$ flips between cell types; oxidation is always at the anode.
- Convert time to **seconds** in $Q = It$; use the correct **electron ratio**.
- Include **water** as a competitor in aqueous electrolysis.
- Always **check charge balance** in half-equations.
- Positive E°_{cell} means **feasible**, not fast.

Glossary

Term	Definition
Redox reaction	Electron-transfer reaction (simultaneous oxidation + reduction).
Oxidation / Reduction	Loss / gain of electrons (ox. number increases / decreases).
Oxidising agent	Electron acceptor; is itself reduced.
Reducing agent	Electron donor; is itself oxidised.
Oxidation number	Charge an atom would have if all bonds were ionic.
Half-equation	Shows one process with its electrons.
Conjugate redox pair	Oxidised and reduced forms linked by e^- transfer.
Anode / Cathode	Electrode of oxidation / reduction.
Salt bridge	Maintains neutrality and completes a galvanic circuit.
E° (standard potential)	Reduction potential of a half-cell vs the SHE at standard conditions.
SHE	Standard Hydrogen Electrode; reference defined as 0.00 V.
Electrolysis	Using electrical energy to drive a non-spontaneous redox reaction.
Overpotential	Extra voltage above E° needed to discharge a species (esp. gases).
Faraday constant F	Charge of one mole of electrons = $96\,500\text{ C mol}^{-1}$.
Disproportionation	One element simultaneously oxidised and reduced.

A Appendix — Standard Reduction Potentials (25 °C)

All half-equations written as reductions. Strongest oxidant at top (most positive E°); strongest reductant at bottom (most negative E°). Always use the exact values on your official exam data sheet.

Ordered strongest oxidant (top) → strongest reductant (bottom)

Reduction half-equation	E° (V)	Note
$\text{F}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{F}^-$	+2.87	strongest oxidant
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.77	
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	+1.51	acidified permanganate
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-$	+1.36	
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+1.33	acidified dichromate
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.23	
$\text{Br}_2(\text{l}) + 2\text{e}^- \rightarrow 2\text{Br}^-$	+1.09	
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}(\text{s})$	+0.80	
$\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$	+0.77	

$\text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$	+0.40	
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34	
$\text{Sn}^{4+} + 2\text{e}^- \rightarrow \text{Sn}^{2+}$	+0.15	
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00	SHE reference
$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.13	
$\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn}(\text{s})$	-0.14	
$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$	-0.25	
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.44	
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76	used in galvanising
$2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-$	-0.83	water reduction
$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}(\text{s})$	-1.66	
$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	-2.37	
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71	
$\text{Ca}^{2+} + 2\text{e}^- \rightarrow \text{Ca}(\text{s})$	-2.87	
$\text{K}^+ + \text{e}^- \rightarrow \text{K}(\text{s})$	-2.93	
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.04	strongest reductant

CONSTANTS & FORMULAE

$F = 96\,500 \text{ C mol}^{-1}$ · e^- charge = $1.602 \times 10^{-19} \text{ C}$ · $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ · $V_m = 24.79 \text{ L mol}^{-1}$ (25 °C, 100 kPa).

$Q = It$ · $n(\text{e}^-) = Q/F$ · $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ · $\Delta G^\circ = -nFE^\circ_{\text{cell}}$

End of notes — practise balancing half-equations and Faraday calculations under timed conditions.